



EB-GS3470-00

Evaluation Board User Guide

Revision History

Version	ECO	PCN	Date	Changes and/or Modifications
5	042894	—	August 2018	Added 2. EB-GS3470-00 Software User Guide and 3. Ancillary Data Extraction , and updated Section 1 .
4	041087	—	March 2018	Updated Section 1.1 .
3	038782	—	September 2017	Digital and analog ground pin names updated to CORE_GND and A_GND respectively. Updates to Pin A1, A2, F1, G1 in Figure 4-3 .
2	034215	—	November 2016	Pin A7 changed to DNC in Figure 4-3 .
1	029858	—	March 2016	Updates to Figure 4-1: EB-GS3470-00 Schematic to Figure 4-4: EB-GS3470-00 Schematic 4 .
0	029082	—	January 2016	New document.

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Overview

Together with the EB-GS3470-00 Evaluation Board, this document serves as a guide for evaluating the GS3470. The GS3470 is a Semtech 3Gb/s, HD, SD SDI Receiver. For more information on the GS3470, please refer to the Product Data Sheet (PDS-060500). This document is partitioned into the following sections:

- [EB-GS3470-00 User Guide](#)
- [EB-GS3470-00 Schematics](#)
- [EB-GS3470-00 Board Layout](#)
- [EB-GS3470-00 Bill of Materials \(BOM\)](#)

Figure A below shows a block diagram of the features and the functions of the EB-GS3470-00.

The board includes power supplies, USB control interface, two 3G-SDI inputs, two GS3140 equalizers, a 3G-SDI loop-through output, a GS2988 multi-rate cable driver, a GS3470 Receiver, a parallel video output connector, an audio output with four RCA for AES audio, a 27MHz crystal, DIP switches, and a few status indication LEDs.

The GS3470 will automatically detect the input signal (J6 and J7) as SD-SDI, HD-SDI, or 3G-SDI. The deserialized video is available on the parallel output connector (J1). The extracted audio is available as serial audio or AES on audio headers (J4 and J12). A serial digital loop-through output (J10) is also available.

The EB-GS3470-00 also provides a JTAG interface of the used for boundary scan and a USB interface to control GS3470 and the two GS3140's through the GSPI ports on the devices.

Kit Contents

- Semtech EB-GS3470-00 Evaluation Board
- Semtech AB-GS3470 Adaptor Board
- Micro USB Cable (for GUI communication)
- DC Power Terminal

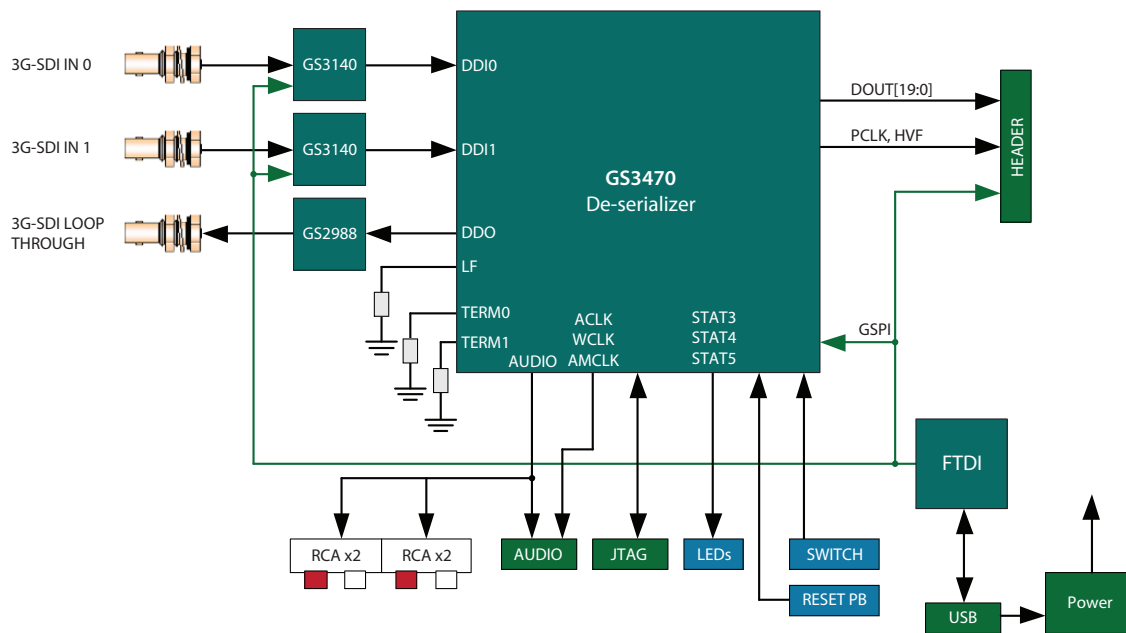


Figure A: Block Diagram

1. EB-GS3470-00 User Guide

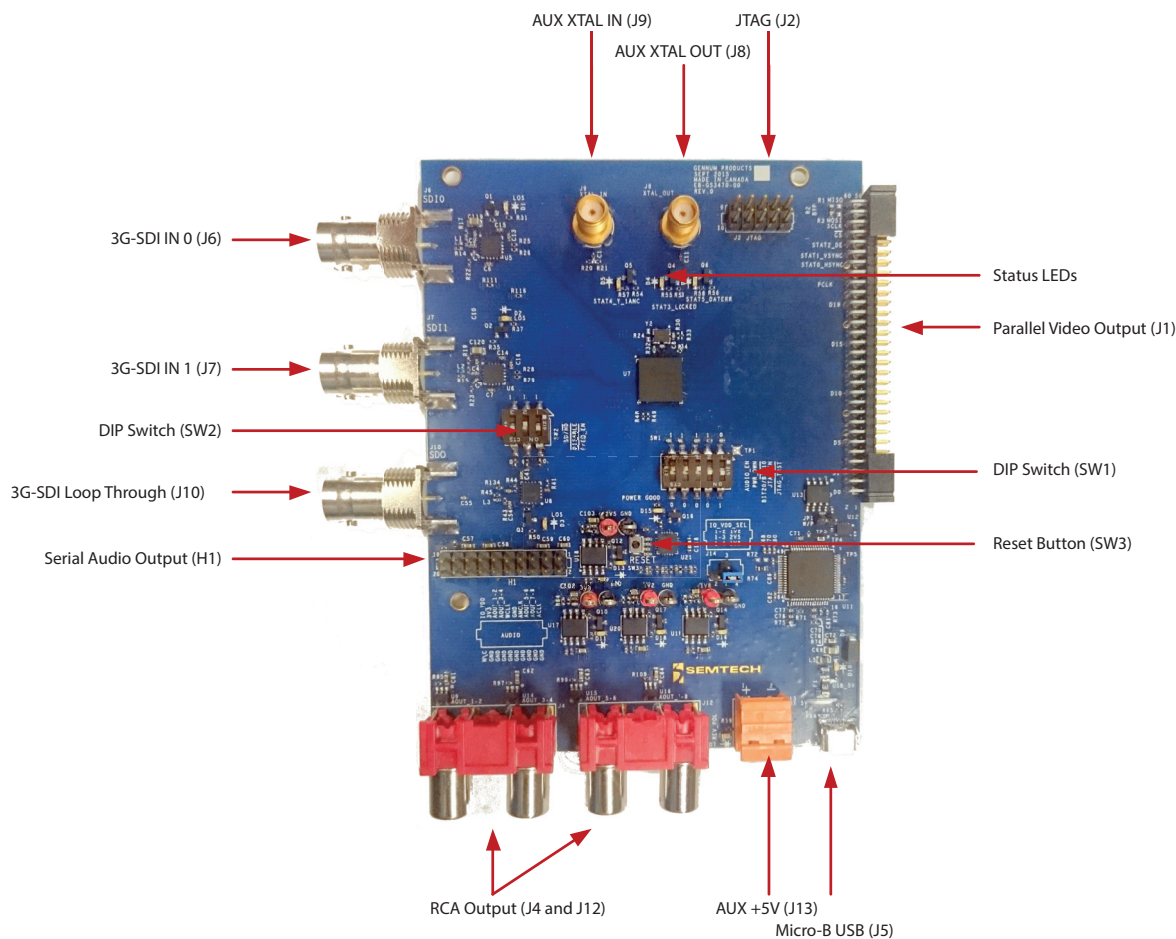


Figure 1-1: GS3470 Evaluation Board (EB-GS3470-00)

1.1 Power (J5 or J13)

The EB-GS3470-00 board is powered by a +5V supply through a Micro-B USB connector (J5) or through the auxiliary power connector (J13). Not all USB ports are capable of supplying enough current to power the board. In this case, use the external +5V auxiliary power connector (J13) with the included DC power terminal. Connect the auxiliary power supply before connecting the USB.

LED D10 is used to indicate if there is voltage present on the +5V input.

LED D7 is used to indicate if the voltage connected to J13 has been reversed.

1.2 SDI Input (J6 and J7)

The EB-GS3470-00 includes two 3G-SDI inputs DC-coupled to the GS3470. The applied video stream will first pass through Semtech's GS3140 Cable Equalizer (PDS-060939) before entering the SDI input pins of the GS3470.

1.3 Switch Settings (SW1 and SW2)

1.3.1 GS3470 Switch Settings (SW1)

Figure 1-2 shows the switch settings for SW1 which control the static configuration input of the GS3470.

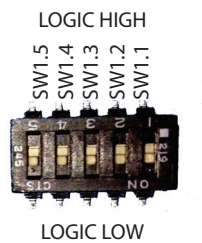


Figure 1-2: GS3470 Switch Configuration

Table 1-1: SW1 Switch Settings

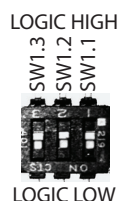
Switch	Pin	Description
SW1.1	JTAG TRST	When HIGH, the GS3470 JTAG interface is in operational mode. When LOW, the GS3470 JTAG interface is held in reset.
SW1.2	JTAG interface enable	When HIGH, the GS3470 JTAG interface is in operational mode. When LOW, the GS3470 JTAG interface is disabled.
SW1.3	BIT20/ $\overline{\text{BIT10}}$	Used to select the video output bus width. When HIGH, 20-bit bus width. When LOW, 10-bit bus width.

Table 1-1: SW1 Switch Settings (Continued)

Switch	Pin	Description
SW1.4	PWR_DWN	When HIGH, places the GS3470 in low-power mode. When LOW, normal operation mode occurs. Note: This switch should be in the HIGH position to allow the GUI to control lower power modes.
SW1.5	Audio_EN/ $\overline{\text{DIS}}$	When HIGH, audio extraction is enabled. When LOW, audio extraction is disabled.

1.3.2 GS2988 Switch Settings (SW2)

Figure 1-3 shows the switch settings for SW2 which controls the static configuration input pins of the GS2988.

**Figure 1-3: GS2988 Switch Configuration****Table 1-2: SW2 Switch Settings**

Switch	Pin	Description
SW1.1	$\overline{\text{EQ_EN}}$	When HIGH, trace-equalization is turned OFF. When LOW, trace-equalization is turned ON.
SW1.2	$\overline{\text{DIS}}$	Sets the GS2988 into low power mode. When HIGH, the GS2988 will be in normal operation mode. When LOW, the entire device is powered down.
SW1.3	SD/ $\overline{\text{HD}}$	When HIGH, the serial data output will meet the SMPTE 259M rise/fall time specification. When LOW, the serial output will meet the SMPTE 292M and SMPTE 424M rise/fall time specification.

1.4 3G-SDI Loop Through (J10)

The output connector J10 is used for the 3G-SDI loop through output. The output for the EB-GS3470-00 evaluation board is capable of the following standards:

- SMPTE 424M
- SMPTE 292M
- SMPTE 259M

The EB-GS3470-00 evaluation board supports DVB-ASI at 270Mb/s and data rates from 270Mb/s to 2.97Gb/s. The output is configured for a single ended coaxial cable terminated with a BNC connection.

1.5 Video Header (J1)

Parallel video data with the PCLK and timing signals H, V, and F are available on a 60-pin connector (J1). Using AB-GS3470-00 will allow the EB-GS3470-00 to interface to legacy evaluation boards.

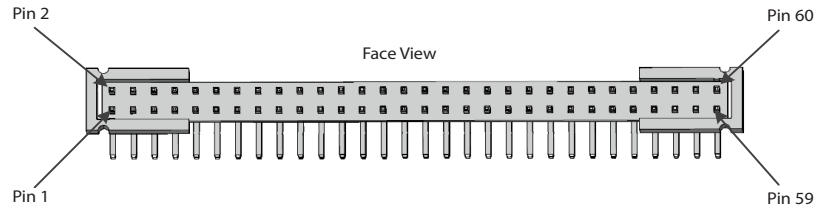


Figure 1-4: Video Header

Table 1-3: Pin Out Video Header

Pin Number	Net Name
J1.1, J1.3, J1.5, J1.7, J1.9, J1.11, J1.13, J1.15, J1.17, J1.19, J1.21, J1.23, J1.25, J1.27, J1.29, J1.31, J1.33, J1.35, J1.37, J1.39, J1.41, J1.43, J1.45, J1.47, J1.49, J1.51, J1.53, J1.55	CORE_GND
J1.2	DATA[0]
J1.4	DATA[1]
J1.6	DATA[2]
J1.8	DATA[3]
J1.10	DATA[4]
J1.12	DATA[5]
J1.14	DATA[6]

Table 1-3: Pin Out Video Header (Continued)

Pin Number	Net Name
J1.16	DATA[7]
J1.18	DATA[8]
J1.20	DATA[9]
J1.22	DATA[10]
J1.24	DATA[11]
J1.26	DATA[12]
J1.28	DATA[13]
J1.30	DATA[14]
J1.32	DATA[15]
J1.34	DATA[16]
J1.36	DATA[17]
J1.38	DATA[18]
J1.40	DATA[19]
J1.42, J1.46	N/C
J1.48	STAT0_HSYNC
J1.50	STAT1_HSYNC
J1.52	STAT2_DE
J1.54	GSPI_CS_N
J1.56	GSPI_SCLK
J1.57	USB_5V_D
J1.58	GSPI_SDIN_EXT
J1.59	USB_GPIO0
J1.60	GSPI_SDOUT_EXT

1.6 Audio Header (H1, J4 and J12)

Audio Outputs (H1 and RCA connectors J4 and J12), are provided to monitor audio in raw form (H1) and AES mode (J4 and J12).

1.6.1 Audio Header (H1)

This connector provides extracted 8-channel audio in conjunction with AMCLK, ACLK, and WCLK.

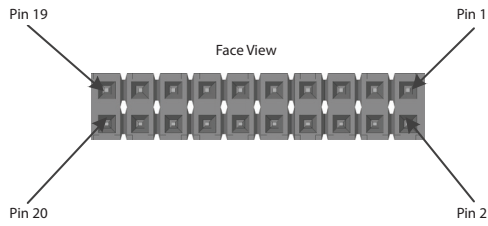


Figure 1-5: Audio Header (H1)

Table 1-4: Pin-Out Audio Header

Pin Number	Net Name
H1.1, H1.3, H1.13, H1.15	AOUT_7_8, AOUT_5_6, AOUT_3_4, AOUT_1_2
H1.2, H1.4, H1.6, H1.8, H1.9, H1.10, H1.12, H1.14, H1.16, H1.18	CORE_GND
H1.5	ACLK
H1.7	AMCLK
H1.11	WCLK
H1.17	+3.3V
H1.19	IO_VDD
H1.20	N/C

This pin-out takes precedence over the silk-screening on the EB-GS3470-00.

1.6.2 AES Model — RCA Connectors (J4 and J12)

For AES mode, AES-encoded CMOS-level signals are distributed through line drivers, and supplied to two RCA connectors (J4 and J12).

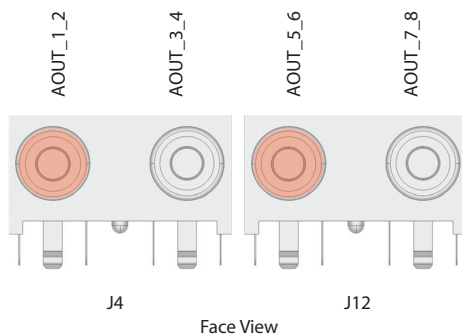


Figure 1-6: RCA Connectors (J4 and J12)

1.7 JTAG Header

The JTAG interface is used for boundary scan testing for the GS3470.

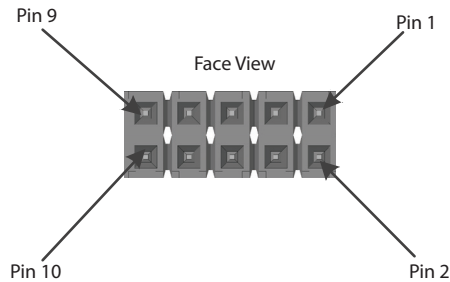


Figure 1-7: JTAG Header (J2)

Table 1-5: Pin-Out JTAG Header

Pin Number	Net Name
J2.1	JTAG_TCK
J2.2, J2.10	GND
J2.3	JTAG_TDO
J2.4	IO_VDD
J2.5	JTAG_TMS
J2.6, J2.7, J2.8	N/C
J2.9	JTAG_TDI

1.8 Status LED's

The functions for the LEDs found on the EB-GS3470-00 are described in [Table 1-6](#).

Table 1-6: Status LEDs Description

Ref	Colour	Programmable	Device	Description
D1	Red	No	GS3140	When active, input Serial Data CH0 signal loss detected.
D2	Red	No	GS3140	When active, input Serial Data CH1 signal loss detected.
D3	Red	No	GS2988	When active, output Serial Data signal loss detected.
D4	Green	Yes	GS3470	When reclocker lock, PLL locked, LED is active.
D5	Green	Yes	GS3470	SD/ $\overline{HD}$, when SD SDI is detected, LED is active.

Table 1-6: Status LEDs Description (Continued)

Ref	Colour	Programmable	Device	Description
D6	Green	Yes	GS3470	When DATA ERROR is detected, LED is inactive.
D7	Red	No	—	Reverse polarity. When polarity is reversed on J13, LED is active.
D10	Green	No	—	When power is present on the 5V input J5 and J13, LED is active.
D11	Green	No	—	When internal +3.3V is good, LED is active.
D13	Green	No	—	When internal +2.5V is good, LED is active.
D15	Green	No	—	Reset indicator, When LED active EB-GS3470-00 is not in reset.
D16	Green	No	—	When internal +1.8V is good, LED is active.
D18	Green	No	—	When internal +1.2V is good, LED is active.

1.9 Modes of Operation

The GS3470 supports four distinct modes of operation that can be accessed through the GUI or through the GSPI interface to internal registers. These modes are: SMPTE mode, Data-Through mode, DVB-ASI mode, and Standby mode.

The GS3470 can automatically detect the format of the incoming signal. If the format of the incoming signal is not recognized, the GS3470 defaults into SMPTE Bypass mode. Alternatively, the device may be forced into any of the four modes through the GUI.

In SMPTE mode, the GS3470 performs full SMPTE processing, and features a number of signal integrity checks and measurement capabilities.

In DVB-ASI mode, 8b/10b decoding is applied to the received data stream.

In Data-Through mode, all forms of SMPTE and DVB-ASI decoding are disabled, and the device can be used as a simple serial-to-parallel converter without incurring the latency penalties associated with SMPTE or DVB-ASI mode.

The device can also operate in a several lower power modes:

- Sleep
- Standby
- Standby with PCLK
- Sleep with SDO
- Standby with SDO Retimed
- Standby with SDO with PCLK
- Standby with SDO Retimed with PCLK

2. EB-GS3470-00 Software User Guide

This guide provides instructions for using the software associated with the EB-GS3470-00.

2.1 Software Setup Instructions

Upon execution of the setup file, the following window will appear as shown in [Figure 2-1](#). To install the software, click next and follow the steps listed in the setup window.

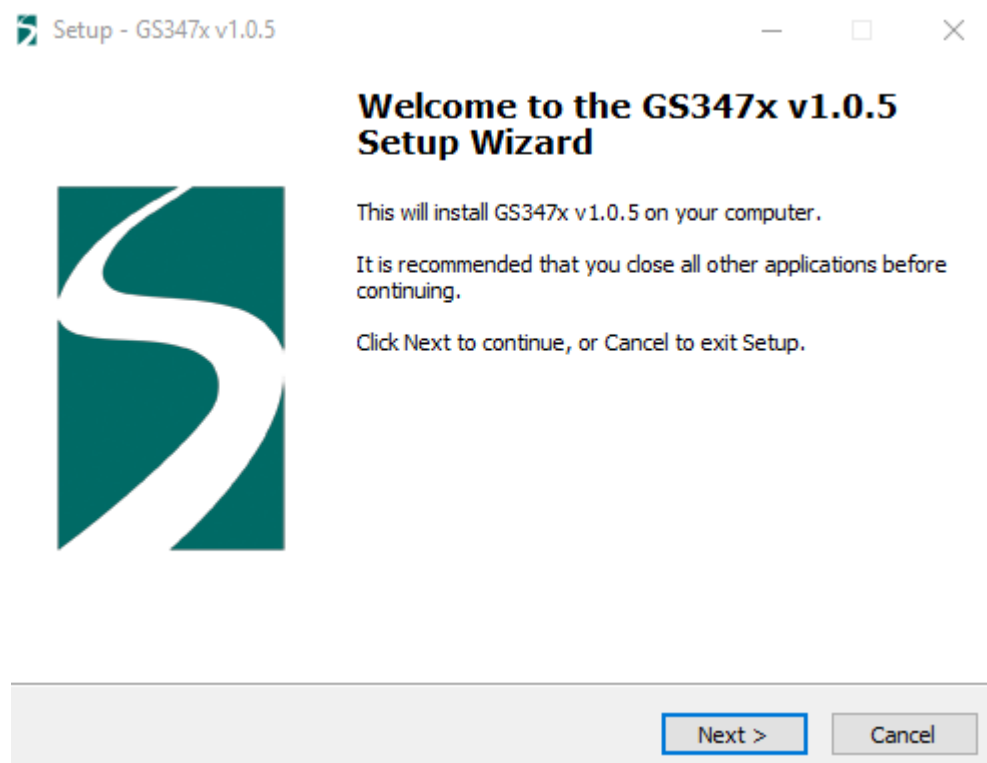


Figure 2-1: EB-GS3470-00 Software Installation Window

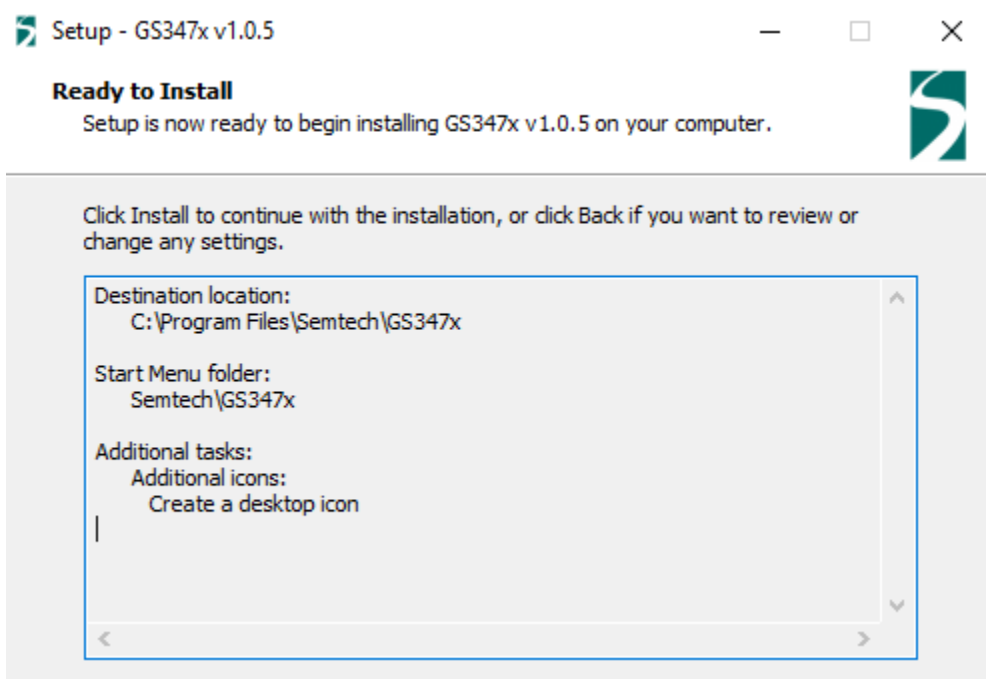


Figure 2-2: EB-GS3470-00 Software Installation Review

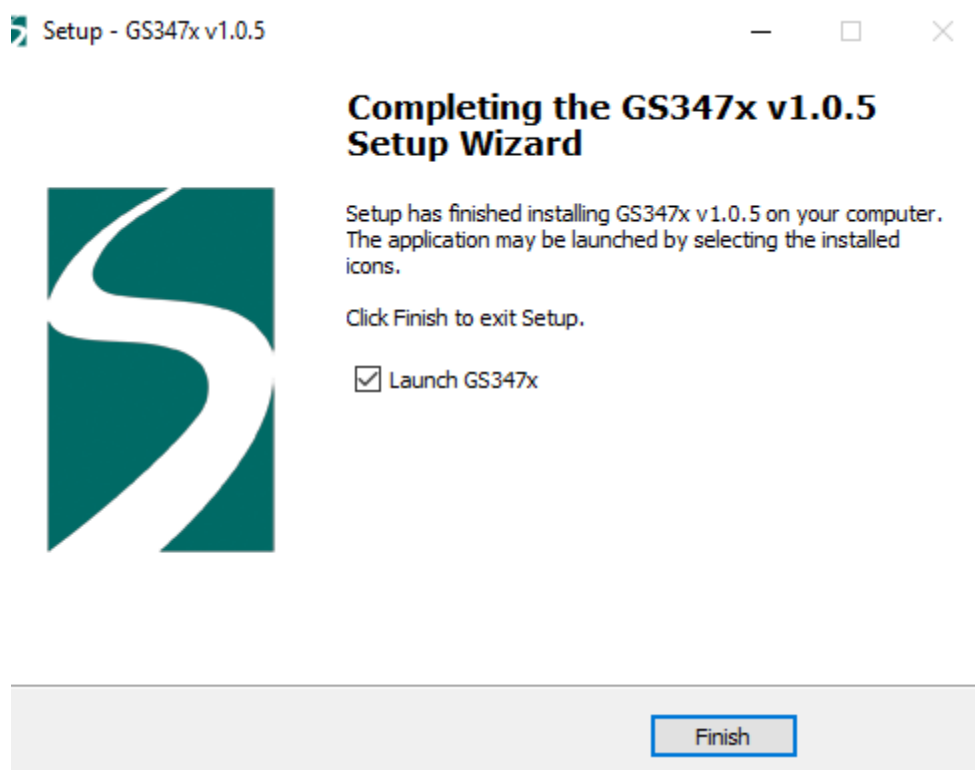


Figure 2-3: EB-GS3470-00 Software Installation Completion

2.2 FTDI CDM Drivers

2.2.1 Error: FTDI Dongle Drivers Missing!

Upon running the GUI software for the first time an error window may arise as shown in Figure 2-4. This error appears when the FTDI CDM Drivers are not installed on the PC.

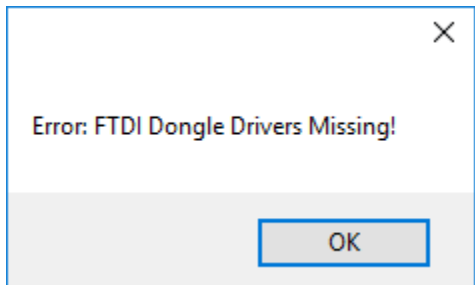


Figure 2-4: Error: FTDI Dongle Drivers Missing!

2.2.2 FTDI CDM Driver Installation

The FTDI CDM Drivers must be downloaded and installed prior to the GUI software operation. The steps are outlined below.

Follow the link to <http://www.ftdichip.com/Drivers/D2XX.htm> for access to a table of D2XX drivers. The first row of the table will contain the drivers required for Windows 10, and subsequent rows for other versions of Windows. Clicking "Setup Executable" in the comments column or clicking http://www.ftdichip.com/Drivers/CDM/CDM21228_Setup.zip and running the executable file will begin the process.

The first step is unpacking the driver package. Click "Extract" to continue.



Figure 2-5: Unpacking Driver Package

Accept the agreement and click “Next” to continue.

Device Driver Installation Wizard

License Agreement



 To continue, accept the following license agreement. To read the entire agreement, use the scroll bar or press the Page Down key.

IMPORTANT NOTICE: PLEASE READ CAREFULLY BEFORE INSTALLING THE RELEVANT SOFTWARE:

This licence agreement (Licence) is a legal agreement between you (Licensee or you) and Future Technology Devices International Limited of 2 Seaward Place, Centurion Business Park, Glasgow G41 1HH, Scotland (UK Company Number SC136640) (Licensor or we) for use of driver software provided by the Licensor(Software).

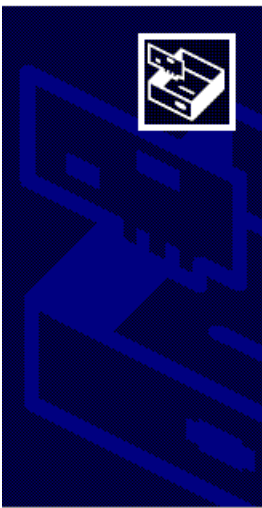
BY INSTALLING OR USING THIS SOFTWARE YOU AGREE TO THE

☒ I accept this agreement ☐ I don't accept this agreement

Figure 2-6: License Agreement

Click “Finish” to close the installation wizard.

Device Driver Installation Wizard



Completing the Device Driver Installation Wizard

The drivers were successfully installed on this computer.

You can now connect your device to this computer. If your device came with instructions, please read them first.

Driver Name	Status
✓ FTDI CDM Driver Packa...	Ready to use
✓ FTDI CDM Driver Packa...	Ready to use

Figure 2-7: Completing the Device Driver Installation Wizard

2.3 GUI Initial Software State

The default state when the software is loaded is shown in [Figure 2-8](#).

Note: Images of the GUI were captured **without** the device connected to the PC.

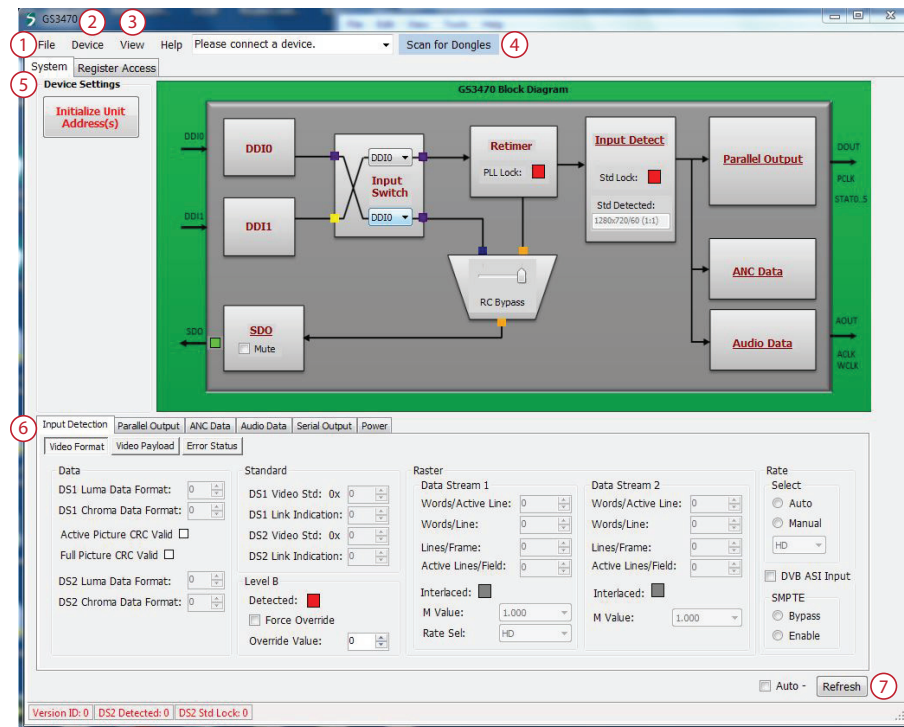


Figure 2-8: System Tab

1. File: use to load/save external configuration.
2. Device: use the drop down menu to choose GS3470Multi rate Deserializer or GS3471 Multi rate Equalizing SDI Receiver.
3. View: use to enable/disable logging window.
4. Dongle Scan: use to select the dongle to enable communication.
5. System Tab (see [Section 2.4](#)) and Register Access Tab (see [Section 2.5](#)).
6. System Tab sub-tabs: [Section 2.4.1](#) through [Section 2.4.6](#).
7. Auto Check Box and Refresh Button: use to auto refresh or manually refresh the GUI state.

2.4 System Tab

The following section will outline the functionality of the sub-tabs contained within the System Tab of the GS3470 GUI.

2.4.1 Input Detection

The Input Detection Tab has three sub-tabs: Video Format (see [Figure 2-8](#)), Video Payload (see [Figure 2-9](#)) and Error Status.

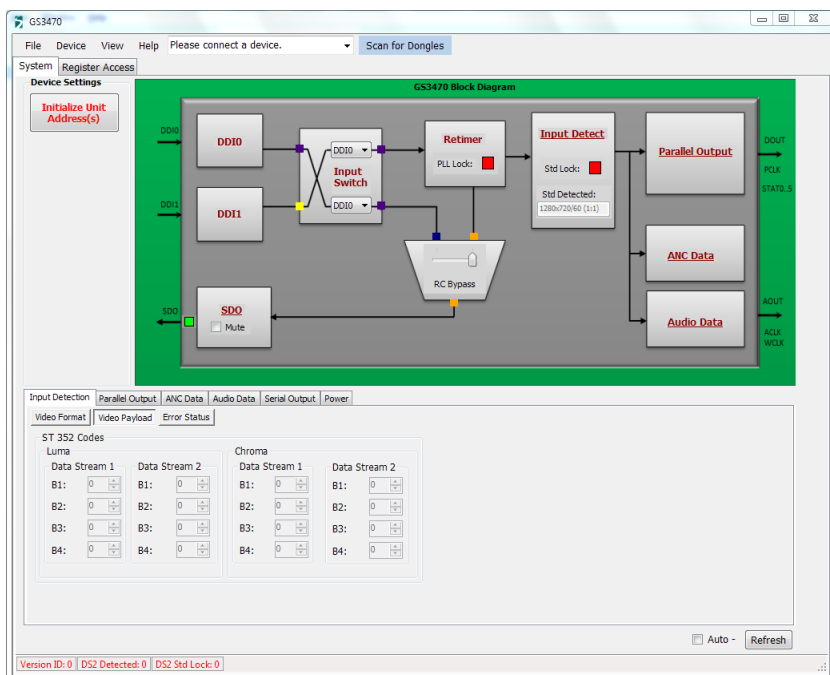


Figure 2-9: Input Detection — Video Payload

2.4.2 Parallel Output

The Parallel Output Tab has three sub-tabs: Processing, Configuration (see Figure 2-10) and EDH Flags.

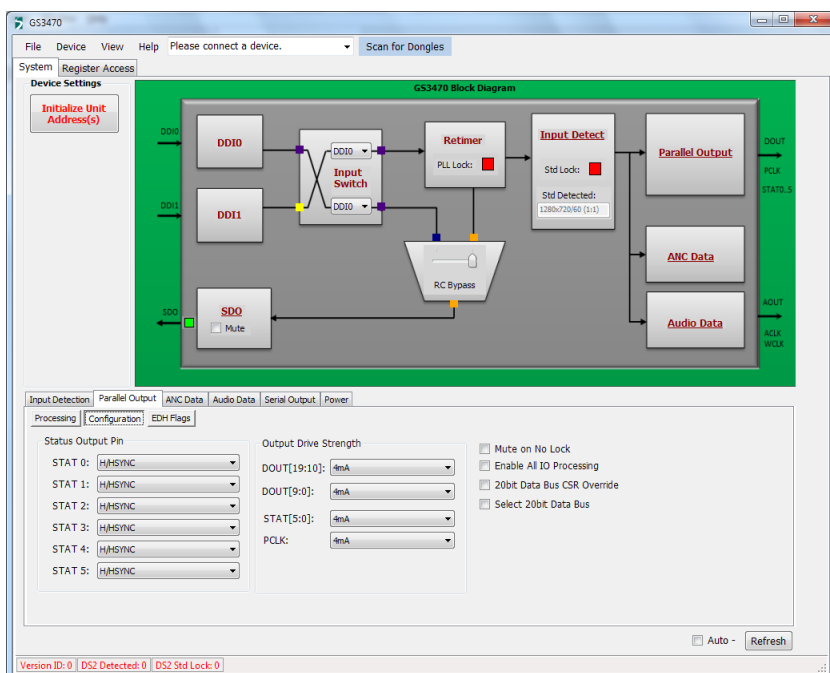


Figure 2-10: Parallel Output — Configuration

2.4.3 ANC Data

The ANC Data Tab has two sub-tabs: Configuration (Figure 2-11), and FIFO (Figure 2-12).

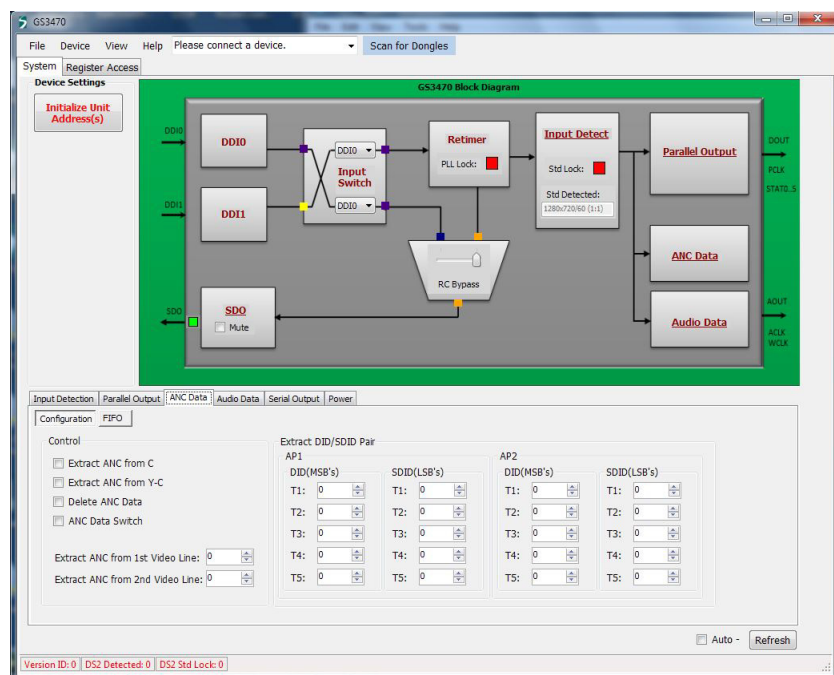


Figure 2-11: ANC Data — Configuration

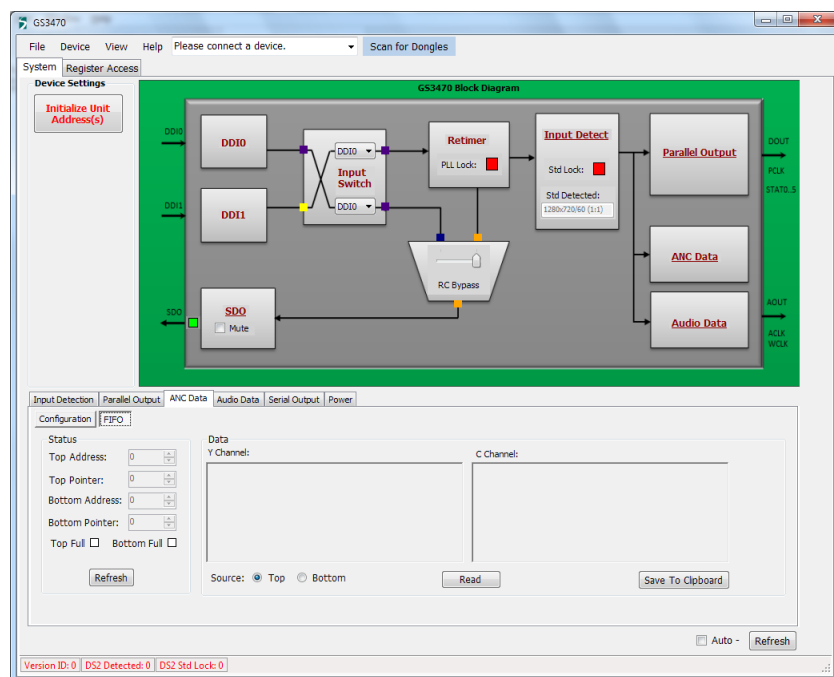


Figure 2-12: ANC Data — FIFO

2.4.4 Audio Data

The Audio Data Tab (see Figure 2-13) has sub-tabs for HD/3G and SD data rates.

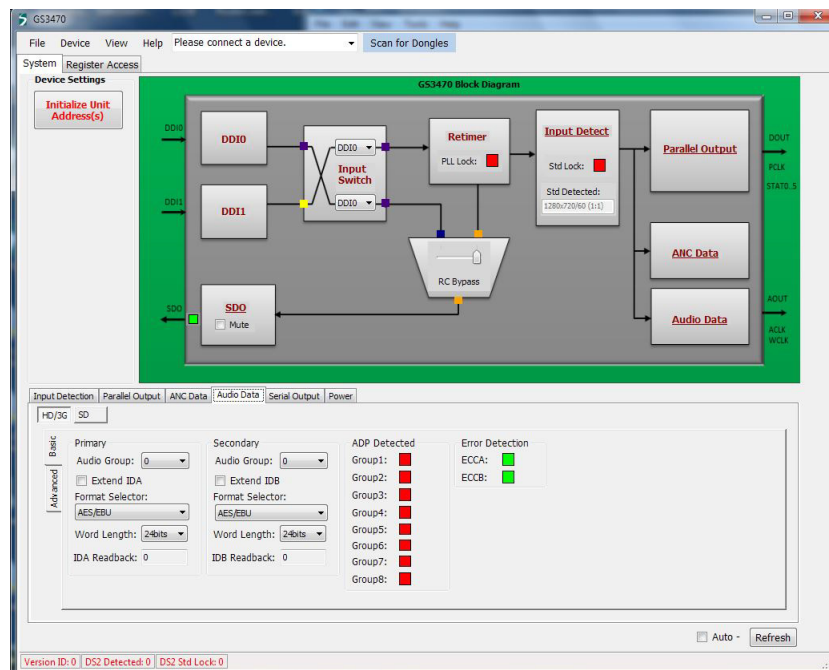


Figure 2-13: Audio Data — HD/3G

2.4.5 Serial Output

The Serial Output Tab is shown in Figure 2-14.

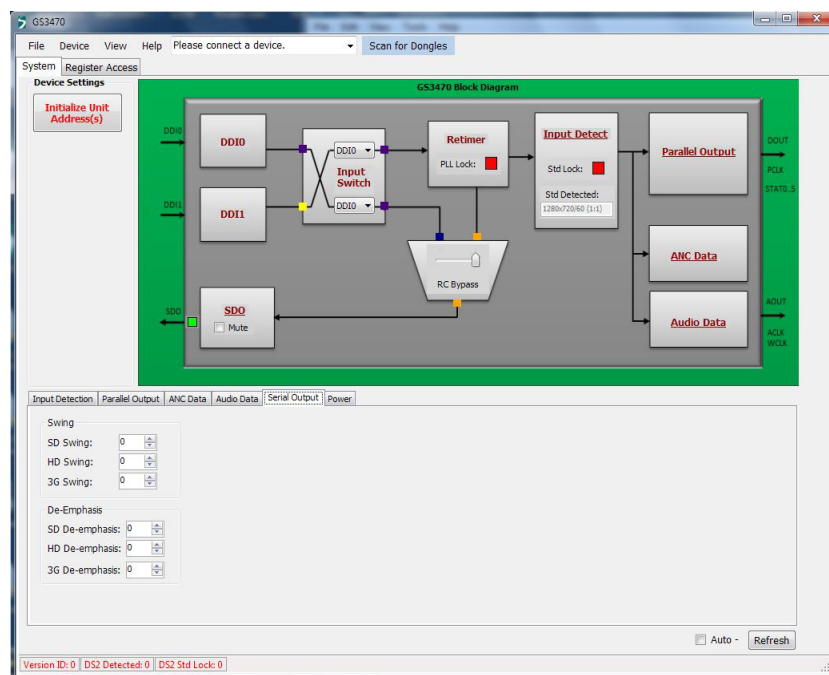


Figure 2-14: Serial Output

2.4.6 Power

The Power Tab is shown in Figure 2-15.

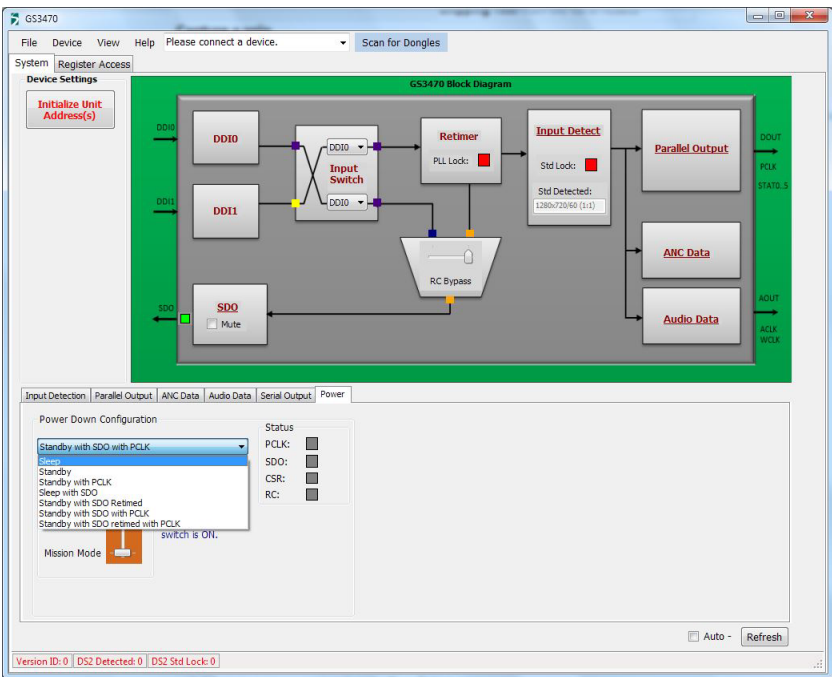


Figure 2-15: Power Tab

2.5 Register Access Tab

Register Access provides manual reading/modifying registers of the GS3470. Please refer to the GS3470 Data Sheet (PDS - 060498).

The screenshot shows the GS3470 software interface with the 'Register Access' tab selected. The main window is divided into two panes. The left pane, titled 'Register Bank', contains a table of registers. The right pane, titled 'Register Access', contains a 'Register Offset' section with a dropdown menu for 'Video', 'HD-Audio', and 'SD-Audio'. Below this is a 'Data' section with a 'Parameter' dropdown and a 'Data' input field. The 'Operations' section contains 'Write', 'Read', and 'Read All' buttons. The 'Configuration' section contains 'Load' and 'Save' buttons. The bottom status bar shows 'Version ID: 0 | DS2 Detected: 0 | DS2 Std Lock: 0'.

OFFSET	CSR_NAME	PARAM_NAME	BITS	MODE	RESETd	DATAb	DATAg	DESCRIPTION
0000	0000	TOPROC_1	TRIS_INS_DS1_MASK	6:0	RW	0000	0000	When high,
0000	0000	TOPROC_1	LNLM_INS_DS1_MASK	1:1	RW	0000	0000	When high,
0000	0000	TOPROC_1	CRC_INS_DS1_MASK	2:2	RW	0000	0000	Disables ins
0000	0000	TOPROC_1	ANC_CHECKSUM_INSERTION_DS1_MASK	3:3	RW	0000	0000	When high,
0000	0000	TOPROC_1	ILLEGAL_WORD_REMAP_DS1_MASK	4:4	RW	0000	0000	When high,
0000	0000	TOPROC_1	TIMING_B61	6:6	RW	0000	0000	0 = Digital f
0000	0000	TOPROC_1	AUD_EXT_MASK	7:7	RW	0000	0000	When high,
0000	0000	TOPROC_1	ANC_DATA_EXT_MASK	8:8	RW	0000	0000	When high,
0000	0000	TOPROC_1	H_CONFIG	9:9	RW	0000	0000	Chooses H
0000	0000	TOPROC_1	EDH_CRC_INS_MASK	10:10	RW	0000	0000	When high,
0000	0000	TOPROC_1	EDH_FLAG_UPDATE_MASK	11:11	RW	0000	0000	When high,
0000	0000	TOPROC_1	TRIS_VIVORD_REMAP_FE1_DISABLE_MASK	12:12	RW	0000	0000	When high,
0000	0000	TOPROC_1	FW_SWITCH_LINE_OVERRIDE_FE1	13:13	RW	0000	0000	If asserted
0000	0000	TOPROC_1	LOW_LATENCY_BYPASS	14:14	RW	0000	0000	When low,
0001	0001	TOPROC_2	TRIS_INS_DS2_MASK	0:0	RW	0000	0000	When high,
0001	0001	TOPROC_2	LNLM_INS_DS2_MASK	1:1	RW	0000	0000	When high,
0001	0001	TOPROC_2	CRC_INS_DS2_MASK	2:2	RW	0000	0000	When high,
0001	0001	TOPROC_2	ANC_CHECKSUM_INSERTION_DS2_MASK	3:3	RW	0000	0000	When high,
0001	0001	TOPROC_2	ILLEGAL_WORD_REMAP_DS2_MASK	4:4	RW	0000	0000	When high,
0001	0001	TOPROC_2	AUDIO_SEL_DS2_DS1B	6:6	RW	0000	0000	Selects dat
0001	0001	TOPROC_2	ANC_EXT_SEL_DS2_DS1B	7:7	RW	0000	0000	Selects dat
0001	0001	TOPROC_2	ANC_EXT_SEL_DS1_Y_DS2_Y	8:8	RW	0001	0000	In 3G level
0001	0001	TOPROC_2	DS_SWAP_3G	9:9	RW	0000	0000	Swaps ds1
0001	0001	TOPROC_2	REGEN_3S2_MASK	10:10	RW	0000	0000	When high,
0001	0001	TOPROC_2	TRIS_VIVORD_REMAP_FE2_DISABLE_MASK	11:11	RW	0000	0000	When high,
0001	0001	TOPROC_2	FW_SWITCH_LINE_OVERRIDE_FE2	12:12	RW	0000	0000	If asserted
0001	0001	TOPROC_2	DSB_AUDET	13:13	RW	0001	0000	THIS NEED

Figure 2-16: Register Access — GS347x CSR

The GS347x Memory Tab is shown in Figure 2-17.

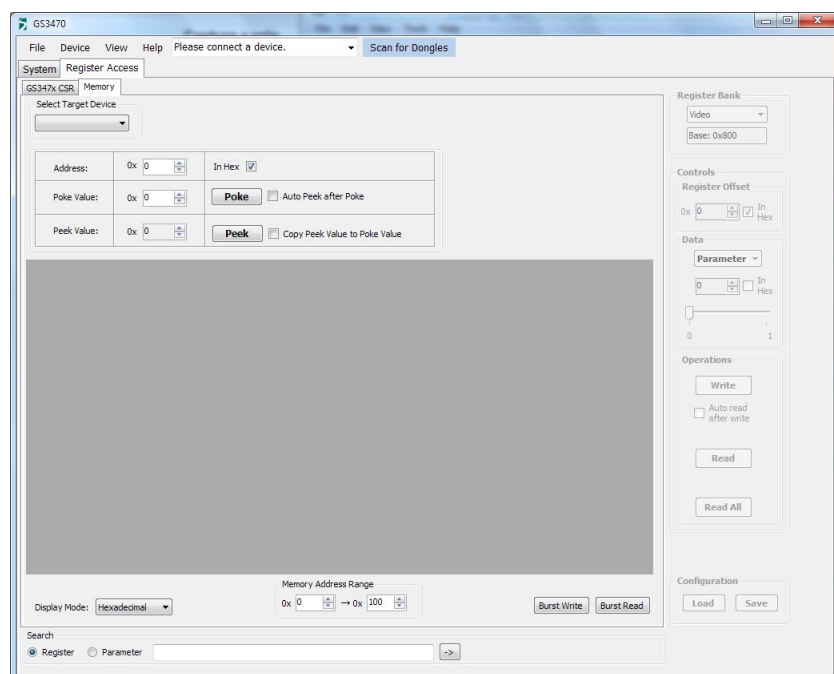


Figure 2-17: Register Access — Memory

3. Ancillary Data Extraction

This procedure uses the ancillary data extraction features of the GS3470 to extract embedded packets. The receivers have two ANC packet FIFOs so data may be read from one FIFO as the other is filling with data. The two FIFOs have the same logical address, and only one may be accessed at a time.

Table 3-1: Required Registers

Register Name	Address	Bits
ANC_LINEA	818	10:0
ANC_LINEB	819	10:0
ANC_TYPE_1_DS2	823	15:0
ANC_DATA_SWITCH	817	3:3
HD_ANC_Y1_C2	817	1:1
HD_ANC_C2	817	0:0
ANC_FIFO	C00 to FFF*	15:0

*Two banks with the same address.

Steps:

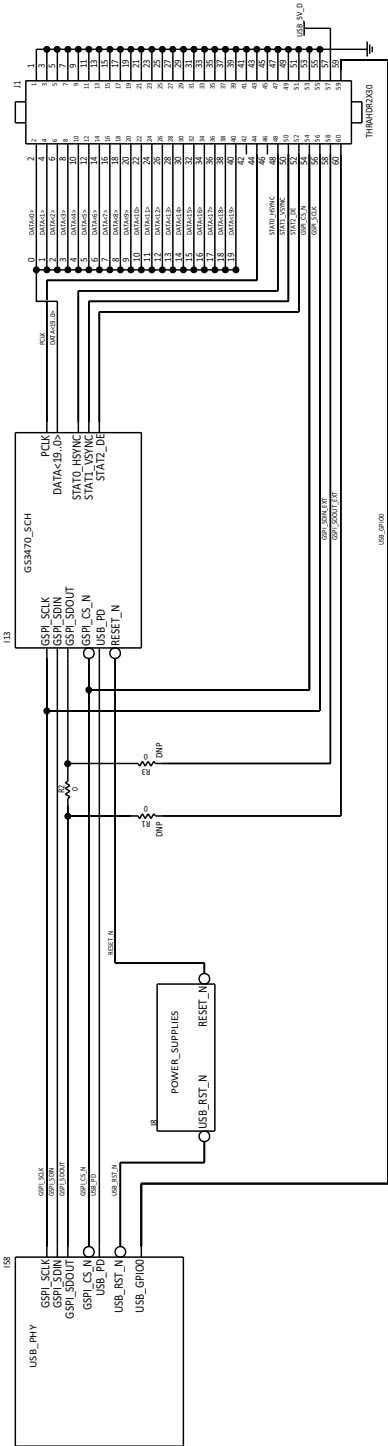
1. ANC_LINEA programmed with line number of ANC packet. ANC_LINEB may also be programmed for interlaced formats.
Note: Defining the line numbers to extract ANC packet is the optional setting. If leaving two line number registers to zero, the GS3470 will search for the matched ANC packets in the whole frame; otherwise, it will search for the matched ANC packets in the specific line(s).
2. ANC_TYPE_1_DS2 programmed with DID and SDID of ANC packet.
3. ANC_DATA_SWITCH set HIGH.
4. At this step ANC data can be extracted from Y (DS1) only, C (DS2) only, or from both (see [Table 3-2](#)). Once the selection is made the FIFO begins to fill.

Table 3-2: Extraction Selection

ANC Extraction Source	HD_ANC_C2	HD_ANC_Y1_C2
Y Only	LOW	LOW
C Only	HIGH	LOW
Y and C	LOW	HIGH

5. Wait for packet to be written into Bank 1.
6. ANC_DATA_SWITCH set LOW (moves write pointer to Bank 2).
7. Read ANC_FIFO (read ANC packet from Bank 1).
8. ANC_DATA_SWITCH set HIGH (moves read point to Bank 2, clears Bank 1)
9. Wait for packet to be written into Bank 2.
10. ANC_DATA_SWITCH set LOW (moves write pointer to Bank 1).
11. Read ANC_FIFO (read ANC packet from Bank 2).
12. ANC_DATA_SWITCH set HIGH (moves read pointer to Bank 1, clears Bank 2).
13. Go to 5.

4. EB-GS3470-00 Schematics



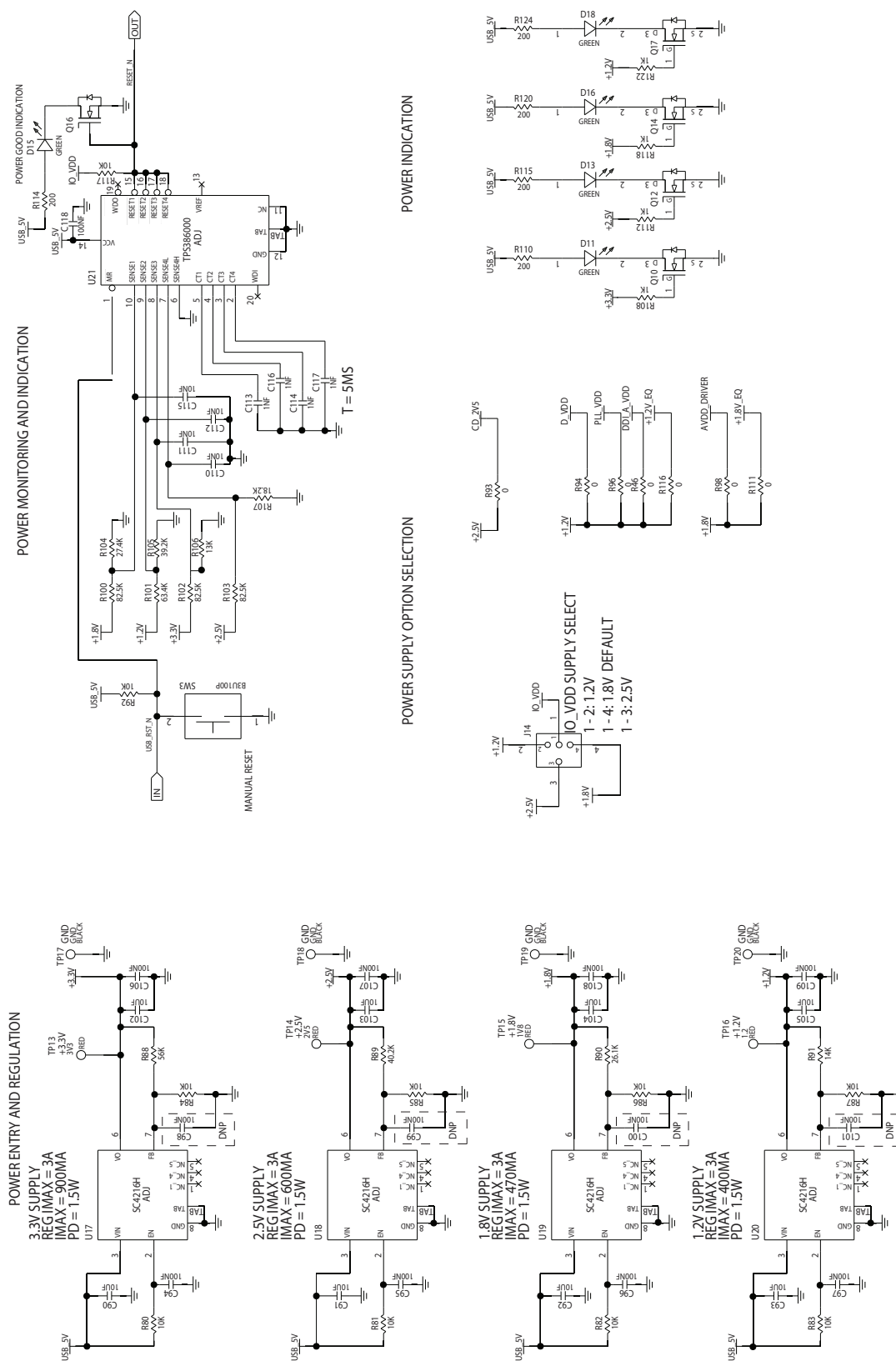
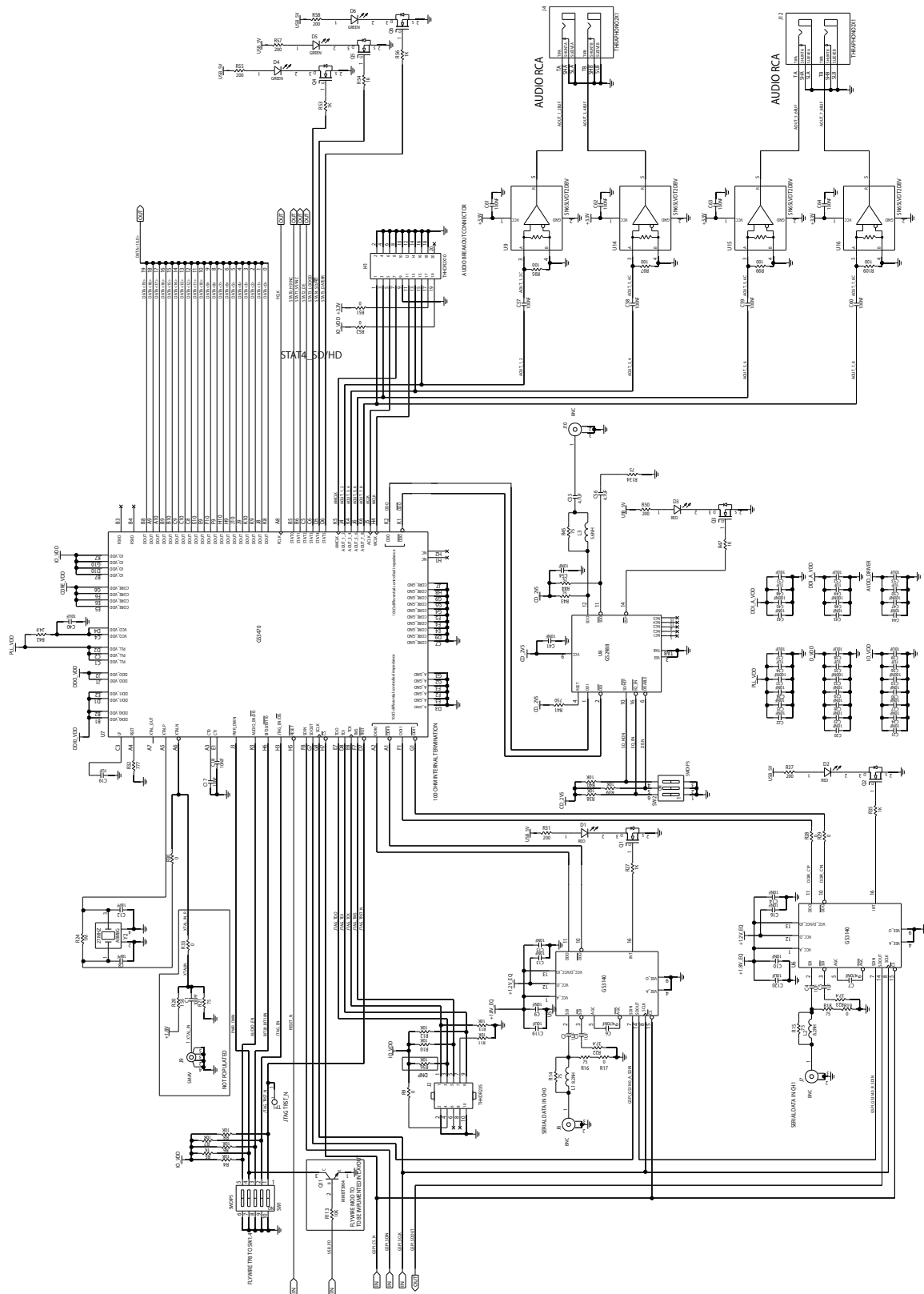
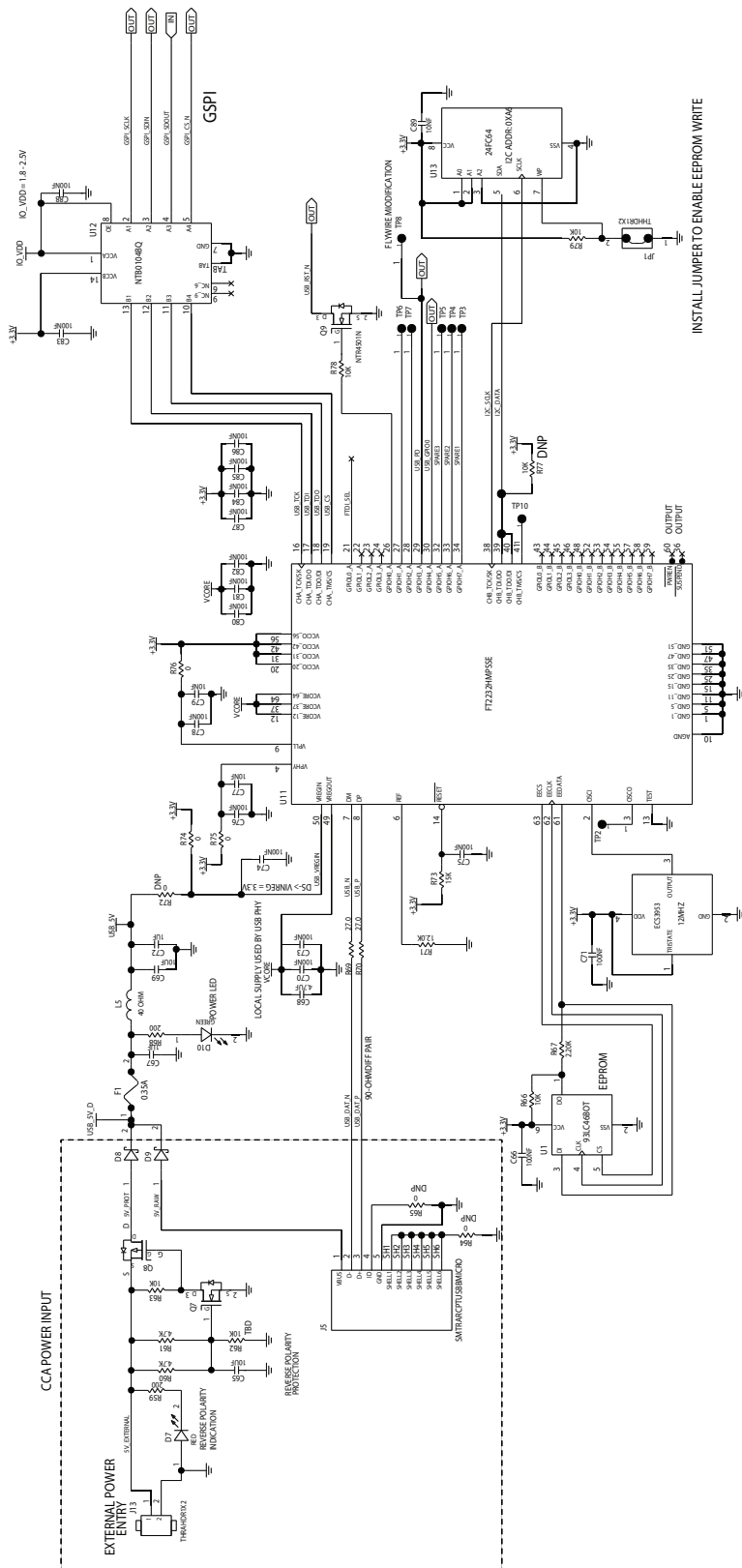


Figure 4-2: EB-GS3470-00 Schematic 2



MODIFICATION CAN BE MADE TO ALLOW AUTO SELECT HD SD

Figure 4-3: EB-GS3470-00 Schematic 3



USB PHY FIELD IN RESET FOR 0.4 - 1.5 MS DURING POWER UP

Figure 4-4: EB-GS3470-00 Schematic 4

5. EB-GS3470-00 Board Layout

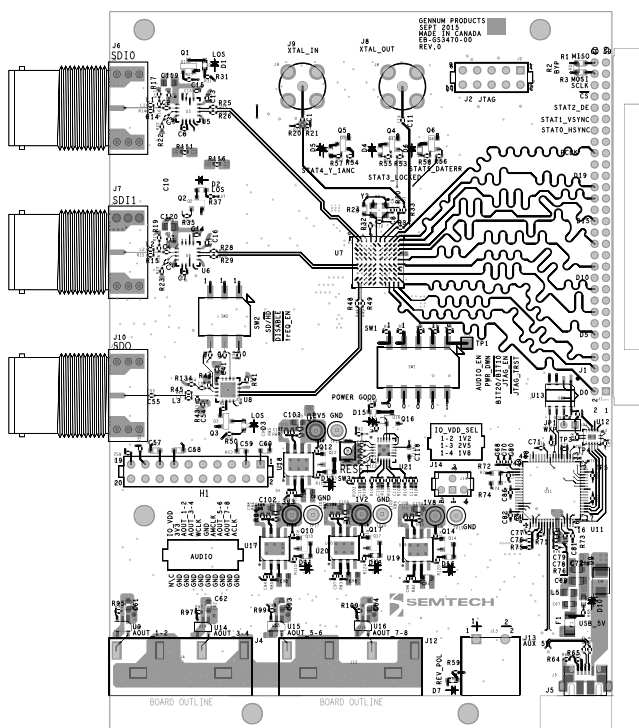


Figure 5-1: Layer 1 (TOP)

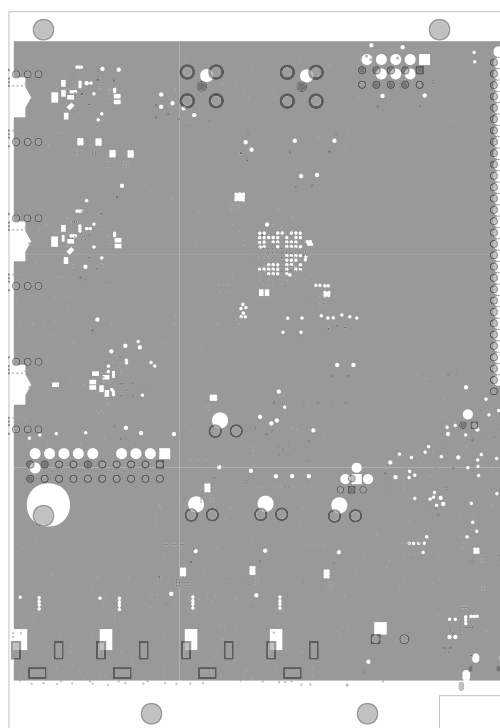


Figure 5-2: Layer 2 (GND1)

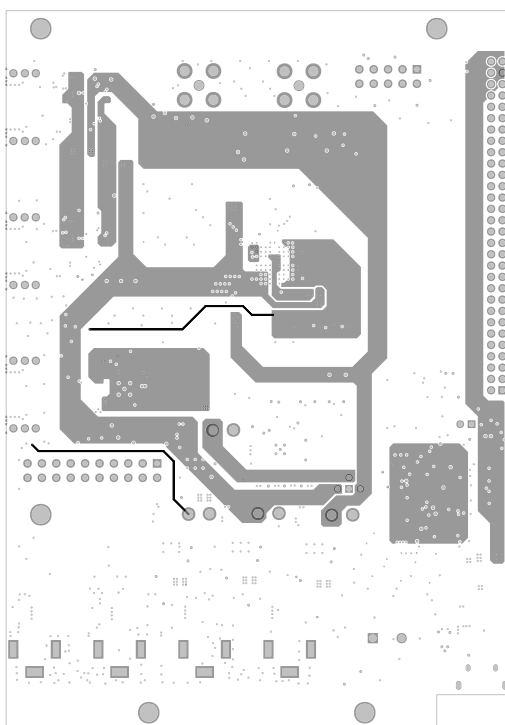


Figure 5-3: Layer 3 (PWR1)

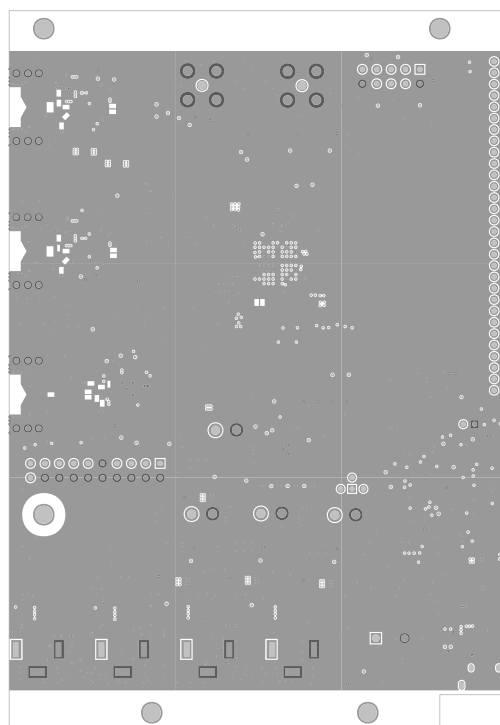


Figure 5-4: Layer 4 (GND2)

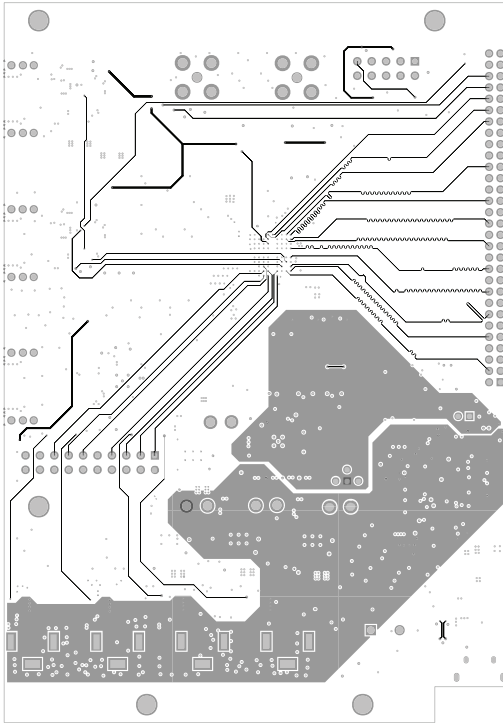


Figure 5-5: Layer 5 (SIG1)

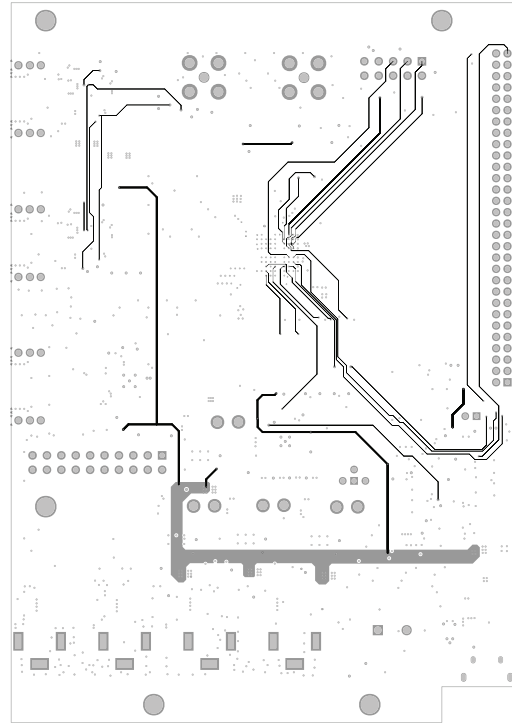


Figure 5-6: Layer 6 (SIG2)

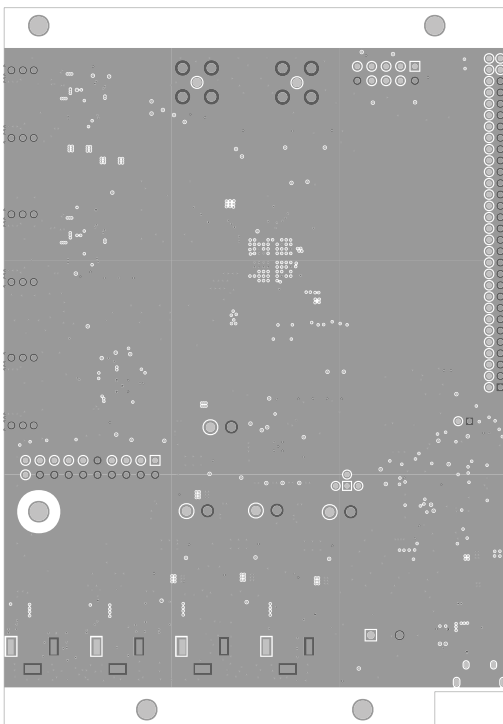


Figure 5-7: Layer 7 (GND3)

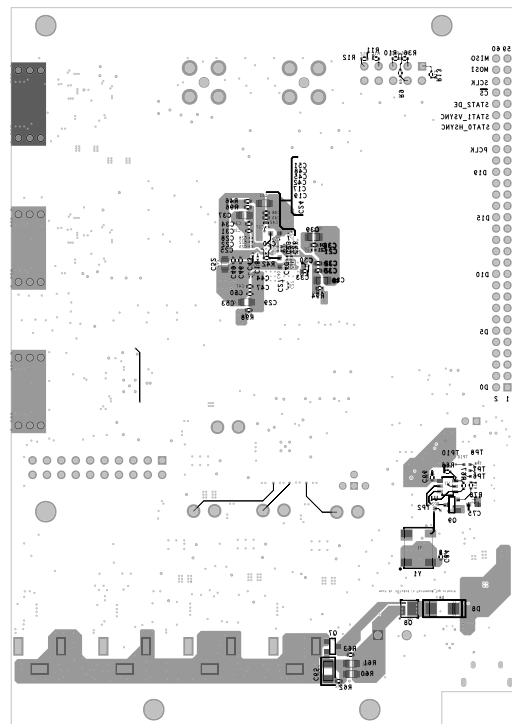


Figure 5-8: Layer 8 (BOT)

6. EB-GS3470-00 Bill of Materials (BOM)

Table 6-1: Bill of Materials

Quantity	Ref Description	Description
4	C2, C3, C4, C5	Capacitor; Ceramic 1.0μF 10V 10% X5R 0402
2	C6, C7	Capacitor; Ceramic 0.47μF 10V 10% X5R 0402
2	C8, C12	Capacitor; Ceramic RF 18pF 50V 5% C0G 0402
15	C9, C10, C13, C14, C15, C16, C41, C54, C77, C79, C89, C110, C111, C112, C115	Capacitor; Ceramic 10000pF 25V 10% X7R 0402
16	C17, C18, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C42, C43, C44	Capacitor; Ceramic 10000pF 16V 10% X7R 0201
3	C19, C67, C72	Capacitor; Ceramic 1.0μF 25V 10% X5R 0603
30	C31, C32, C33, C45, C46, C47, C66, C70, C71, C73, C74, C76, C78, C80, C81, C82, C83, C84, C85, C86, C87, C88, C94, C95, C96, C97, C106, C107, C108, C109	Capacitor; Ceramic 0.1μF 16V 10% X7R 0402
6	C34, C35, C36, C48, C49, C50	Capacitor; Ceramic 1.0μF 6.3V 10% X7R 0402
9	C37, C38, C39, C51, C52, C53, C69, C119, C120	Capacitor; Ceramic 10μF 16V C5R 20% X5R 0805
9	C40, C90, C91, C92, C93, C102, C103, C104, C105	Capacitor; Ceramic 10μF 6.3V 20% X5R 0603
3	C55, C56, C68	Capacitor; Ceramic 4.7μF 6.3V 20% X5R 0402
10	C57, C58, C59, C60, C61, C62, C63, C64, C75, C118	Capacitor; Ceramic 0.1μF 25V 10% X7R 0603
1	C65	Capacitor; Ceramic 10μF 25V X5R 1206
4	C113, C114, C116, C117	Capacitor; Ceramic 1000pF 50V 10% X7R 0402
3	D1, D2, D3	LED; Red TSS Type SMD
9	D4, D5, D6, D10, D11, D13, D15, D16, D18	LED; 1.6x0.8mm 568nm Green CLR SMD
1	D7	LED; 1.6x0.8mm 625nm Red CLR SMD
2	D8, D9	Diode; Schottky 20V 3A SMD
1	F1	Fuse; PTC Reset 6V 0.35A 1206
1	H1	Connector; HDR Drkaway 0.100 78-position VERT (20 POS USED)
1	J1	Connector; Header 3x30-position 2mm RA ES
1	J2	Connector; Header 10-position 0.100DL Gold
2	J4, J12	Connector; RCA Phono 2P R/A Phono Jack
1	J5	Connector; receptacle REV Micro USB Type B

Table 6-1: Bill of Materials (Continued)

Quantity	Ref Description	Description
3	J6, J7, J10	Connector; Bulkhead/PCB edge mount BNC 75Ω
1	J13	Connector; Header 5.08MM 2POS 90 ORG
1	J14	Connector; Header 2mm dual STR 72-position (6 positions used, 2 pins removed)
1	JP1	Connector; Header 2mm single STR 36-position (2 positions used)
2	L1, L2	Inductor; 8.2nH 1500mA ±2% 0402
1	L3	Inductor; 5.6nH 300mA 0402
1	L5	Ferrite; 1.5A 40Ω 0805 SMD
13	Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q9, Q10, Q12, Q14, Q16, Q17	Transistor; MOSFET N-CH 20V 3.2A SOT23-3
1	Q8	Transistor; MOSFET P-CH 30V 11A 8-PQFN
1	Q11	Transistor; GP NPN AMP SOT23-3
27	R4, R6, R7, R8, R10, R11, R12, R13, R38, R39, R40, R62, R63, R66, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R92, R113, R117	Resistor; 10.0kΩ 1/16W 1% 0402 SMD
2	R2, R74	Resistor; 0.0Ω 1/10W 5% 0603 SMD
1	R5	Resistor; 1.0kΩ 1/16W 1% 0402 SMD
21	R9, R17, R19, R25, R26, R28, R29, R30, R46, R48, R49, R51, R52, R75, R76, R93, R94, R96, R98, R111, R116	Resistor; 0.0Ω 1/16W 5% 0402 SMD
8	R14, R15, R16, R18, R43, R44, R45, R134	Resistor; 75.0Ω 1/16W 1% 0402 SMD
2	R22, R23	Resistor; 37.4Ω 1/16W 1% 0402 SMD
10	R27, R35, R47, R53, R54, R56, R108, R112, R118, R122	Resistor; 1.00kΩ 1/16W 1% 0402 SMD
13	R31, R37, R50, R55, R57, R58, R59, R68, R110, R114, R115, R120, R124	Resistor; 200Ω 1/16W 1% 0402 SMD
1	R32	Resistor; 777Ω 1/16W 0.1% 0402 SMD
1	R41	Resistor; 750Ω 1/16W 1% 0402 SMD
1	R42	Resistor; 24.9Ω 1/10W 1% 0402 SMD
2	R60, R61	Resistor; 4.70kΩ 1/4W 1% 0805 SMD
1	R67	Resistor; 2.20kΩ 1/10W 1% 0402 SMD
2	R69, R70	Resistor; 27.0Ω 1/10W 1% 0402 SMD
1	R71	Resistor; 12.0kΩ 1/10W 1% 0402 SMD
1	R73	Resistor; 15.0kΩ 1/16W 1% 0402 SMD

Table 6-1: Bill of Materials (Continued)

Quantity	Ref Description	Description
1	R88	Resistor; 56.0kΩ 1/10W 1% 0402 SMD
1	R89	Resistor; 40.2kΩ 1/10W 1% 0402 SMD
1	R90	Resistor; 26.1kΩ 1/10W 1% 0402 SMD
1	R91	Resistor; 14.0kΩ 1/16W 1% 0402 SMD
4	R95, R97, R99, R109	Resistor; 100Ω 1/16W 1% 0402 SMD
3	R100, R102, R103	Resistor; 82.5kΩ 1/16W 1% 0402 SMD
1	R101	Resistor; 63.4kΩ 1/16W 1% 0402 SMD
1	R104	Resistor; 27.4kΩ 1/16W 1% 0402 SMD
1	R105	Resistor; 39.2kΩ 1/16W 1% 0402 SMD
1	R106	Resistor; 13.0kΩ 1/16W 1% 0402 SMD
1	R107	Resistor; 18.2kΩ 1/16W 1% 0402 SMD
1	SW1	Switch; Tape Seal 5 POS SMD
1	SW2	Switch; Tape Seal 3 POS SMD
1	SW3	Switch; Tact SPST Without Ground SMD MOM 153GF
4	TP13, TP14, TP15, TP16	Test Point; PC Multi-purpose Red
4	TP17, TP18, TP19, TP20	Test Point; PC Multi-purpose Black
1	U1	IC; EEPROM 1kBIT 2MHz SOT23-6
2	U5, U6	IC; Multi-Rate Adaptive 3G SDI Equalizer (Semtech GS3140)
1	U7	IC; 3G, HD, SD SDI Receiver (Semtech GS3470)
1	U8	IC; Dual Slew-Rate, Cable Driver with 3Gb/s Capability (Semtech GS2988)
4	U9, U14, U15, U16	IC; Diff Line Receiver SOT23-5
1	U11	IC; USB HS dual UART/FIFO 64-LQFP
1	U12	IC; TXRX Translating 3STATE 14DHVQFN
1	U13	IC; EEPROM 64kBIT 1MHz 8SOIC
4	U17, U18, U19, U20	IC; Regulator very low input/very low dropout 3A Regulator with enable
1	U21	IC; Supply monitor ADJ open drain/COLL QUAD 20QFN
1	Y1	IC; OSC 12.000MHz 50ppm 3.3V SMD
1	Y2	Crystal; 27.000MHz 18pF 3.2x2.5x1mm SMD



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